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Cyano group modified g-C₃N₄: Molten salt method achievement and promoted photocatalytic nitrogen fixation activity

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ABSTRACT

Photocatalytic nitrogen fixation is one of the most green and sustainable methods for NH₃ production under ambient conditions. Exploration of high-efficiency photocatalysts paves the way for realizing ammonia synthesis under ambient conditions. In this study, through the molten salt method retreatment approach, we design a cyano group modified g-C₃N₄ (CG-CN), which enables high-efficiency visible-light-driven nitrogen fixation in the absence of any organic scavengers and precious-metal cocatalysts. The cyano group (—C≡N) can promote the separation of photogenerated electron-hole pairs and simultaneously induce the nitrogen activation based on the strong electron-withdrawing character. The introduction of the cyano group results in a 12.8 times improved photocatalytic nitrogen fixation activity in sharp contrast to that of pristine g-C₃N₄ (CN). This work provides a feasible route to modify photocatalysts with functional surface group for nitrogen fixation.

1. Introduction

Ammonia (NH₃), an essential chemical product, has been playing a crucial part in plastic, textile industries, agriculture, and food production [1]. The great demand of NH₃ promotes the research on nitrogen fixation, which converts the ample atmospheric N₂ to the NH₃, but suffering from the difficulty in dissociating the strong nonpolar N≡N covalent triple bond [2]. Currently, the century-old Haber-Bosch still dominate the industrial NH₃ production, which must be performed under harsh reaction condition (300–500 °C, 15–25 MPa) in the presence of iron-based catalyst to surmount the kinetic limitation [3]. As a consequence, the Haber-Bosch process needs to consume a large amount of energy [4]. Aiming at the reduction of pollutants originated from the Haber-Bosch process, an environmentally friendly approach with low energy expenditure for nitrogen fixation is desiderated. Photocatalysis has enormous potential in nitrogen fixation as a green and economical technology with solar energy being the driving force in the reaction process [5].

Photocatalytic nitrogen fixation was first achieved under ultraviolet (UV) irradiation on a Fe-doped TiO₂ powder by Schrauzer and Guth in 1977 [6]. Owing to insufficiency of an efficient method to activate N₂, research on photocatalytic nitrogen fixation was rarely reported before 2015. Then Zhang et al reported the BiOBr surface modified by oxygen vacancy had the ability to activate N₂, which attracted extensive

attention of investigators [7]. Recently, Ye and co-workers reported the OV in ultrafine Bi₅O₇Br nanotubes for boosting the solar-driven nitrogen fixation [8]. Zhang reported the severely distorted structure and compressive strain in the CuCr-LDH nanosheets to enhance N₂ chemisorption [9]. Shiraishi reported the Ti³⁺ species on the oxygen vacancies could serve as nitrogen adsorption sites [10]. Xiong and co-workers reported Mo species played a significant role in refining the defect states of W₁₈O₄₉ to facilitate N₂ activation and dissociation [11]. However, most of the past researches mainly relied on metal-based photocatalyst, while the employment of metal-free catalysts was less investigated.

Graphitic carbon nitride (g-C₃N₄), has recently attracted widespread attention on account of the advantages of rich raw material, low cost, simple synthesis and visible light response [12]. Although g-C₃N₄ has superiority in many aspects, it also suffers from the narrow light absorption range, limited carrier separation efficiency and insufficient active sites [13]. Thus, researchers have developed extensive strategies to promote the photocatalytic behavior of g-C₃N₄, such as nanostructure engineering [14], doping with heteroatoms to enhance light absorption and improve the separation efficiency of photo-generated electron-hole pairs [15], being exfoliated to a few layers [16], or constructing heterostructure [17]. Recently, surface modification has received extensive attention as it can offer the coordinatively unsaturated sites to activate N₂ [18]. Some researchers explored different kinds of

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methods to obtain surface modified g-C₃N₄ for nitrogen fixation, such as ultrathin amine-functionalized g-C₃N₄ [19], Ga₂O₃-DBD/g-C₃N₄ [20], and SiW₁₂/K-C₃N₄ [21]. Our previous study also found that reasonable surface modification could result in binuclear end-on coordinated mode to activate N₂ [22,23]. Inspired by the recent research, fabricating surface modified g-C₃N₄ with well-designed catalytic activation center capable of activating the N≡N triple bond are potentially achievable, which still lacks of exploration.

In this work, we developed a cyano group modified g-C₃N₄ (CG-CN) through molten salt method retreatment approach with the addition of melamine. The cyano group can modulate the electronic structure of g-C₃N₄ due to its strong electron-withdrawing character [24]. Moreover, it can facilitate the separation of photogenerated electron-hole pairs. As expected, such cyano group modified g-C₃N₄ is found to be more efficient in nitrogen activation and results in a 12.8 times enhanced photocatalytic nitrogen fixation activity under visible light. We believe that our findings deepen the comprehension on the surface modification of g-C₃N₄ based material and provide a promising method for the reasonable design of photocatalysts for more efficient nitrogen fixation.

2. Experimental

2.1. Synthesis process

Cyano group modified graphitic carbon nitride (CG-CN) was designed as follows: 5 g melamine was added to an alumina boat and sintered at 600 °C at a heating rate of 5 °C/min for 2 h under N₂ flow. Then, 0.6 g graphitic carbon nitride produced from the previous step mixed with a certain melamine and LiCl/KCl salt. The mixture was transferred into an alumina boat and sintered at 550 °C at the heating rate of 5 °C/min for 2 h in an air atmosphere. The mass ratios of melamine: g-C₃N₄ was 1:10, 3:10, and 5:10 (labeled CG-CN-1, CG-CN-3 and CG-CN-5), respectively. CN was prepared by heating melamine at 600 °C at a heating rate of 5 °C/min for 2 h in a N₂ atmosphere and then calcined at 550 °C at the heating rate of 5 °C/min for 2 h in an air atmosphere.

2.2. Photocatalytic measurements

The photocatalytic nitrogen fixation experiment was carried out in quartz photochemical reactor. 100 mL diluted hydrochloric acid containing 50 mg photocatalyst was magnetic stirred for 30 min in the dark with the high-purity N₂ bubbled in the solution. Then the 300 W Xe lamp (PLS-SXE300/300UV) with a cutoff filter ($\lambda > 400$ nm) was turned on. 2 mL mixture was collected and centrifugalized from the suspension after 1 h irradiation. The concentration of ammonia was measured by the indophenol indicator spectrophotometry method with an ultraviolet-visible spectrophotometer (the details of NH₃/NH₄⁺ concentration analysis are presented in the [Supporting Information](#)).

2.3. Characterization

The crystal structure of samples was characterized using a powder X-ray diffractometer (XRD) with Cu-K α radiation. A field-emission scanning electron microscope (FESEM) was employed to observe the morphology of the samples. Transmission electron microscope (TEM, FEI Tecnai G²) working at 300 kV was employed to characterize the chemical composition of the samples. Ultraviolet-visible diffuse reflectance spectra (UV-vis DRS) were carried out on a HITACHI UH-4150 spectrophotometer. Fourier transform infrared (FTIR) spectroscopy was detected employing IR Affinity-1 FT-IR spectrometer. X-ray photoelectron spectroscopy (XPS) was performed on a Thermo Scientific ESCALAB 250Xi X-ray photoelectron spectrometer with Al K α excitation source (1486.6 eV). The photoluminescence (PL) were carried out using HORIBA FluoroMax-4 spectrometer. Time-resolved PL spectra were measured on a FluoroMax-4 spectrometer. Nitrogen

temperature programmed desorption (N₂-TPD) measurements were performed using the Quantachrome Autosorb-iQ-C chemisorption analyzer. In situ Fourier transform infrared (FT-IR) spectroscopy for the coadsorption of N₂ and H₂O before and after simulated sunlight irradiation was performed on Nicolet IS50 spectrometer (MCT detector).

The Mott-Schottky plots, photocurrent response and electrochemical impedance spectroscopy (EIS) were performed on an Autolab PGSTAT302N (ECO-Chemie) electrochemical workstation with a standard three-electrode system. The Ag/AgCl (saturated KCl) act as the reference electrode and the counter electrode was a Pt flake (the details of photoelectrochemistry tests are presented in the [Supporting Information](#)).

2.4. Density-functional-theory (DFT) calculation

The first-principles calculations were performed based on density functional theory (DFT) by the Cambridge Serial Total Energy Package (CASTEP) in Materials Studio software. The Perdew-Burke-Ernzerh (PBE) with Generalized Gradient Approximation (GGA) was utilized. Self-consistent field (SCF) tolerance was 5.0×10^{-7} eV/atom. A k-points grid of $3 \times 3 \times 2$ was set for relaxation calculations and energy calculations. And 544.2 eV was set as energy cut-off. The energy is less than 0.001 eV/cell for geometry optimization.

The bond energies were calculated as:

$$E_{\text{bond}} = E_{\text{total}} - (E_{\text{C}\equiv\text{N}} + E_{\text{CN}})$$

3. Results and discussion

The cyano group modified graphitic carbon nitride was synthesized by molten salt treatment with melamine as additive ([Fig. 1a](#)). To reveal the difference between CG-CN-3 and CN, the X-ray diffraction (XRD) analysis and Fourier transform infrared (FTIR) was conducted. As shown in [Fig. 1b](#), two characteristic peaks located at 13.3° and 27.3°, which match well with the (1 0 0) and (0 0 2) crystal planes of g-C₃N₄, are visible in CN [25]. In sharp contrast, a typical sample of CG-CN-3 exhibits a weaker and broader diffraction peak at about 12.9°, while the (0 0 2) peak shifts to a larger 2θ value. The obvious change in (1 0 0) peak corresponds to the decrease in the order degree of in-plane structural packing, which might be due to the surface modification of cyano group in the g-C₃N₄ framework [26]. Besides, the XRD pattern of molten salt g-C₃N₄ (MSCN) was measured ([Fig. S1](#)). The (0 0 2) peak for MSCN and CG-CN-3 both display shift to larger 2θ value, indicating that the molten salt process indeed enhances the crystallinity of g-C₃N₄. To mention, the full width at half maximum (FWHM) of (0 0 2) peak for MSCN shows a significant decrease compared with CN and CG-CN-3, suggesting that the crystallinity of CG-CN-3 decreases and the interlayer spacing increases compared with MSCN [27]. Therefore, the shift of (0 0 2) peak was due to the synergistic effect of molten salt and cyano group [28]. To further demonstrate the existence of surface modification of cyano group in CG-CN-3, we conducted the Fourier transform infrared (FTIR). The FTIR spectrum for CG-CN-3 displays three new peaks at ~ 1000 cm⁻¹, ~ 1200 cm⁻¹, 2179 cm⁻¹, corresponding to C–O stretching of C–OH, C–N stretching of N–CN and an asymmetric stretching vibration of cyano group, respectively ([Fig. 1c](#)) [29]. The peaks at ~ 1200 cm⁻¹, 2179 cm⁻¹ declare the successful modification of cyano group in the g-C₃N₄ framework [30]. The peak at ~ 1000 cm⁻¹ emerges after molten salt process, which may be due to surface adsorption of oxhydryl during the washing process. Both CN and CG-CN-3 ([Fig. 1c](#)) show typical peak for the out-of-plane bending mode of heptazine rings at 810 cm⁻¹, characteristic peaks for the “melon” framework located between 1200 cm⁻¹ and 1800 cm⁻¹, the intensity of broad peaks in 3000–3300 cm⁻¹ region corresponding to N–H stretching vibrations exhibit no obvious decrease, which indicates that the terminal amine can be maintained in CG-CN-3. Therefore, no direct bonding between cyano group and the terminal amine is existent [31].

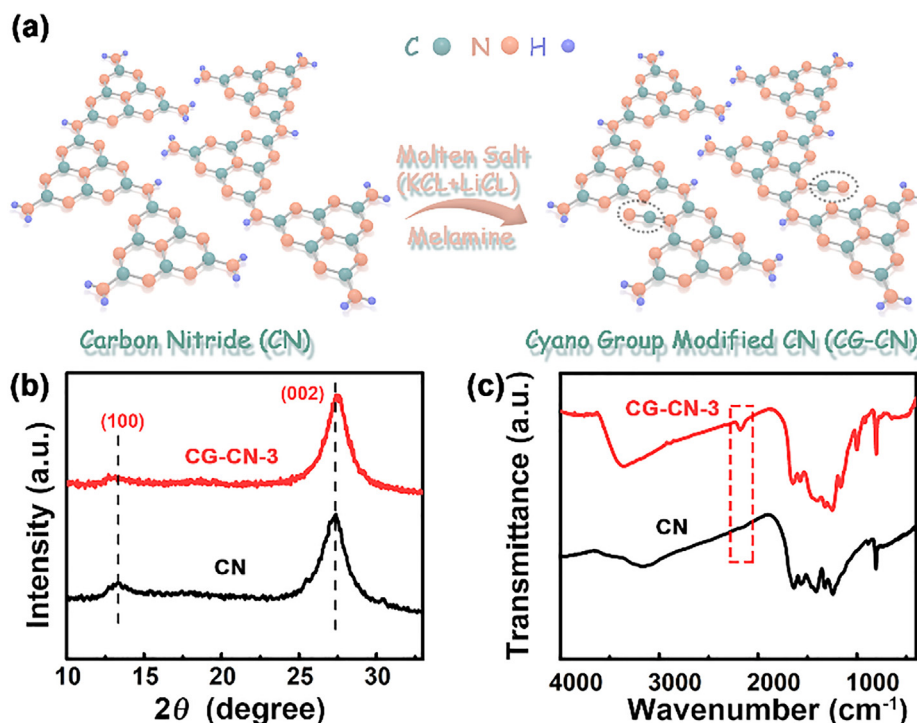


Fig. 1. (a) Schematic for the formation of cyano group modified $g\text{-C}_3\text{N}_4$ (CG-CN), (b) XRD patterns, and (c) FTIR spectra.

The intensity of band peaks 3500 cm^{-1} corresponding to the O–H stretching vibrations show a significant enhancement, which suggests that the hydroxyl group is more likely to adsorb on the surface of CG-CN-3 [28].

Energy dispersive spectrometer (EDS) result of CG-CN-3 (Table S1) show a relatively lower N atomic percent compared with the EDS of CN, consistent with the surface modification of cyano group. To mention, the element contents of K and Cl in CG-CN-3 verge on zero in the error range, indicating no molten salt residue. Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images (Figs. S2, S3) manifest surface modification of cyano group here presents little effect on the morphology. The high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) image and corresponding energy-dispersive X-ray spectroscopy (EDS) elemental mapping exhibit the homogeneous distribution of the C and N, which indicate the “melon” framework displays no significant change.

Fourier transform infrared (FTIR) spectroscopy is performed to confirm the generation process of cyano group (Fig. S4). The CG-CN-3 shows a stronger peak of cyano group than both sample calcined without molten salt (CN-M-3) and sample calcined without melamine (MSCN), which indicates cyano group are manufactured due to the simultaneous presence of melamine and molten salt (Fig. S4a). Molten salt can break hydrogen bonds, which promote the formation of the cyano group [32]. On the other hand, adsorptive melamine work as reagent for generating cyano group. The amount of cyano group can be adjusted by the mass ratio of melamine (Fig. S4b, c). FTIR spectra (Fig. S4d) show that the peak of cyano group reaches the maximum at 550°C with the increase of heating temperature, which suggests the optimum condition is 550°C .

To further investigate the position of cyano group, X-ray photoelectron spectroscopy (XPS) is performed. The C 1s spectra of CG-CN-3 and CN (Fig. 2a) consisted of three contributions at 284.8, 286.4, and 288.2 eV, corresponding to adventitious carbon, $\text{C}-\text{NH}_x$ ($x = 1, 2$) on the edges of heptazine units and $\text{N}-\text{C}\equiv\text{N}$ coordination in the framework of $g\text{-C}_3\text{N}_4$, respectively [33]. The intensity at 286.4 eV of CG-CN-3 is higher than the same feature for CN (Fig. 2a), which can be treated as extra evidence for the formation of cyano group since cyano group

possesses similar C 1s binding energy to $\text{C}-\text{NH}_x$ [34]. The N 1s spectrum for CN (Fig. 2b, top) is composed of three peaks at 401.0, 400.0, and 398.6 eV, assigning to NH_x group in the heptazine framework, tricoordinated ($\text{N}_{3\text{C}}$) and bicoordinated ($\text{N}_{2\text{C}}$) nitrogen atoms, respectively. After the surface modification of cyano group, the $\text{N}_{3\text{C}}$ peak exhibited a shift to higher binding energy (Fig. 2b, bottom). As published in pervious research, the decrease in electron cloud density could weaken the electron screening effect, resulting in a higher binding energy [35]. Therefore, we consider that the higher binding energy shift of $\text{N}_{3\text{C}}$ in CG-CN-3 is due to the strong electron-withdrawing character of cyano group [36].

To mention, the intensity of $\text{N}_{2\text{C}}$ peak has no significant change, suggesting that nitrogen vacancies were not formed on the surface of $g\text{-C}_3\text{N}_4$ [37]. The light absorption band detected by Ultraviolet–visible (UV–vis) diffuse reflectance spectra exhibits no marked alteration, which can be taken as accessional evidence for the inexistence of nitrogen vacancies (Fig. 2c and Fig. S5). Furthermore, the same edge in the XPS valence band of about 2.08 eV and the same flat band potential of CG-CN-3 and CN indicate no significant shift, corresponding to the UV–vis diffuse reflectance spectra (Fig. 2d and Fig. S6). Generally, the photocatalytic nitrogen fixation could occur mainly depend on the position of valence band (VB) and conduction band (CB) in semiconductor [38]. Combined results of XPS valence band with UV–vis diffuse reflectance spectra, the band structure alignments were carried (Fig. S7). Obviously, band gap range of CN and CG-CN-3 can satisfy the thermodynamic reduction potential of N_2 to NH_4^+ (-0.092 eV vs NHE) [39].

Photoluminescence (PL) spectroscopy was employed to explain the impacts of the cyano group on the separation behavior of the photo-generated charge carriers in the photocatalysts. The fluorescence intensity of CG-CN-3 is far below the fluorescence intensity of CN (Fig. 3a). Which suggests the efficient separation of photogenerated charge carriers due to the formation of cyano group [40]. Furthermore, the CG-CN-3 shows a lowest fluorescence intensity compared to the CG-CN-1 and CG-CN-5 (Fig. S8a), which indicates a most efficient photo-generated charge carriers separation in CG-CN-3. The lowest fluorescence intensity appears at the 550°C may be due to an optimum molten

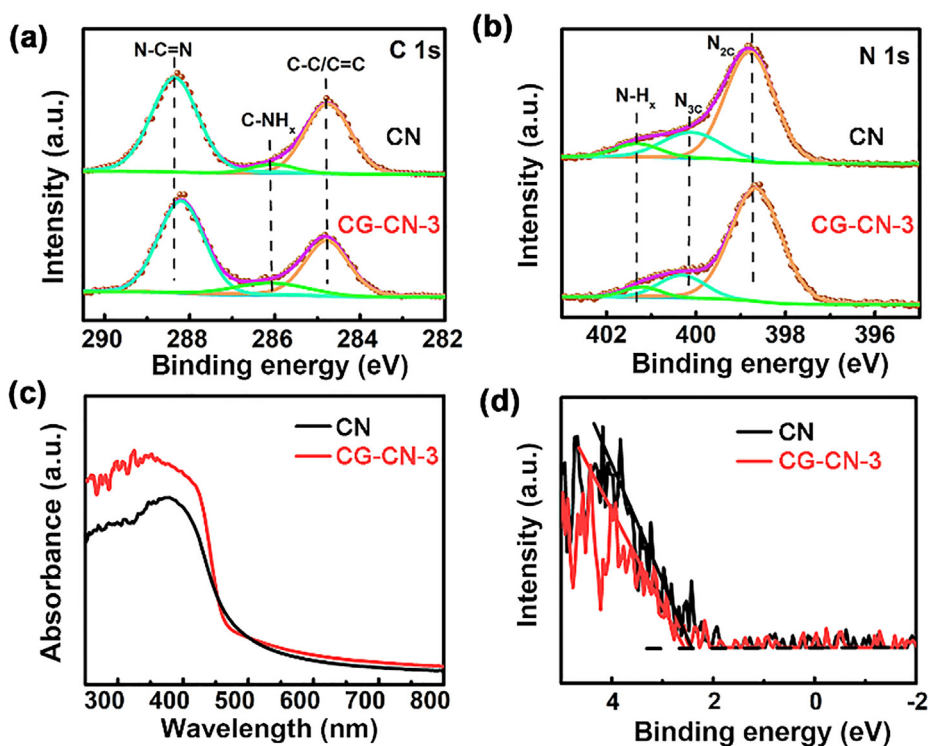


Fig. 2. (a, b) C 1s and N 1s XPS spectra of CN and CG-CN-3, (c) UV-vis diffuse reflectance spectra and (d) VB XPS spectra of CN and CG-CN-3.

salt condition for g-C₃N₄ synthesis (Fig. S8b). According to time-resolved fluorescence decay spectroscopy, the average intensity lifetimes (τ) of CG-CN-3 is 1.85 ns, which is longer than that for CN (1.14 ns) (Fig. 3b). This indicates that the radiative lifetime for charge carriers is prolonged because of introducing cyano group, which should be beneficial for photocatalytic reactions [41].

Influences of charge carriers separation behavior are also carried out by photoelectrochemical measurements (Fig. 3c and d). Transient photocurrent measurements for CN and CG-CN-3 revealed that CG-CN-3 displayed a more sensitive photocurrent response compared with CN during on/off irradiation cycles under visible light (Fig. 3c). The improvement in the photocurrent indicated that separation efficiency of

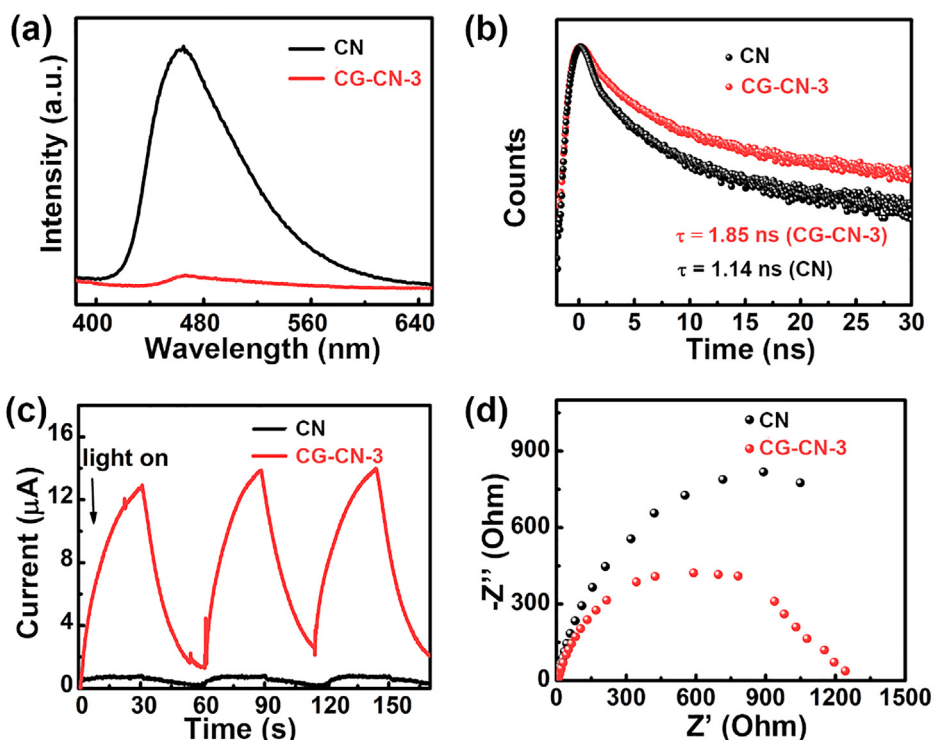


Fig. 3. (a) Photoluminescence spectra, (b) time-resolved fluorescence decay spectra, (c) Transient photocurrent responses under visible light irradiation ($\lambda > 400$ nm) and (d) Electrochemical impedance spectroscopy response of CN and CG-CN-3.

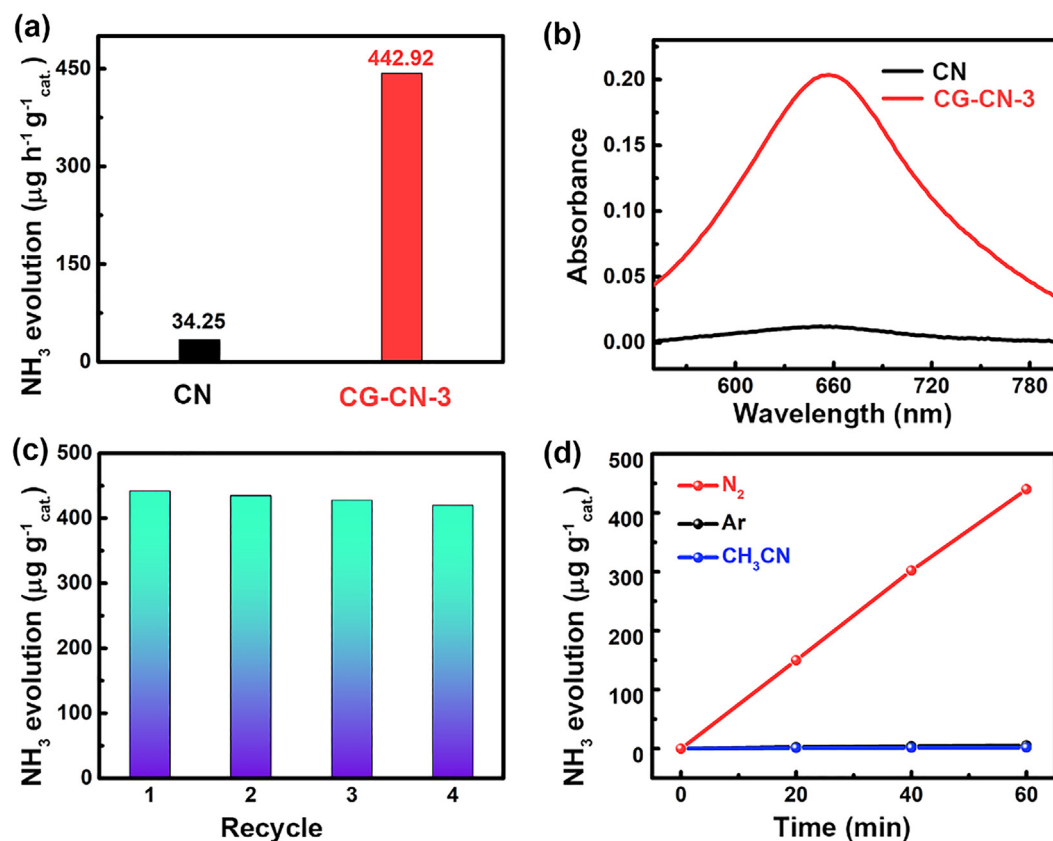


Fig. 4. (a) Nitrogen fixation rate of CN and CG-CN-3 under visible light irradiation ($\lambda > 400$ nm) with no sacrificial agent. (b) Corresponding UV-vis absorption spectra of the electrolyte stained with indophenol indicator. (c) Cycling experimental performance of CG-CN-3. (d) Photocatalytic nitrogen fixation of CG-CN-3 in aprotic solvents (CH_3CN) and under Ar saturation.

the photogenerated charge carriers is obviously improved in the CG-CN-3 [42]. The effect of the cyano group on the kinetics of interfacial charge immigration was measured by EIS analysis. A smaller arc radius in EIS Nyquist curve was in accordance with a more efficient separation of photogenerated electron-hole pairs in semiconductor-based photoelectrodes [43]. As showed in Fig. 3d, the CG-CN-3 showed an obvious decreased radius in the EIS Nyquist curve compared to the CN, suggesting more effective separation and immigration of the photogenerated charge carriers in CG-CN-3, which is consistent with the consequences of the PL tests and photocurrent measurement.

The visible-light photocatalytic nitrogen fixation activities of CN and CG-CN-3 photocatalysts are shown in Fig. 4a and Fig. S9a. The corresponding UV-vis absorption spectra of the product solutions stained with indophenol indicator are shown in Fig. 4b and Fig. S9b. The nitrogen fixation rate of CN is $34.35 \mu\text{g h}^{-1} \text{g}_{\text{cat}}^{-1}$, while CG-CN-3 exhibits nitrogen fixation rate ($442.92 \mu\text{g h}^{-1} \text{g}_{\text{cat}}^{-1}$) (the detecting calibration curve for colorimetric ammonia is shown in Fig. S10). Combined with the FTIR results (Fig. 1c, Fig. S3d and Fig. S11), above C—OH bond almost disappears after 1 h N_2RR (Fig. S11), manifesting it might originate from surface absorbed oxydriyl during the washing molten salt process. In sharp contrast, the surface cyano group can be well maintained after N_2RR , while the photocatalytic activity shows no obvious change (Fig. 4c). These results can exclude the possibility of surface C—OH to serve as a major factor leading to an improvement in nitrogen fixation activity. Therefore, combined with the photocatalytic activity (Fig. 4c and S9), the improvement of nitrogen fixation rate can be ascribed to the formation of cyano group. The CG-CN-3 contained the maximum cyano group exhibited a highest nitrogen fixation rate ($442.92 \mu\text{g h}^{-1} \text{g}_{\text{cat}}^{-1}$), which suggested the increment of cyano group may contribute to the improvement of nitrogen fixation. With increasing the mass ratio of melamine in the molten salt process, the

amount of cyano group first increased and then was almost stable. The improvement of nitrogen fixation rate from CG-CN-1 to CG-CN-3 may result from the increasing content of the cyano group, while the slight diminution of nitrogen fixation from CG-CN-3 to CG-CN-5 may be due to excess melamine which covered the surface defects. Furthermore, the cycling experiment (Fig. 4c) for visible-light photo-catalytic nitrogen fixation over CG-CN-3 was performed. The catalyst can maintain an efficient and durable photocatalytic activity after four repeated runs.

Nitrogen and proton are essential for the nitrogen fixation. In this work the proton from H_2O reacted with nitrogen, which is an energy-saving reaction process compared with those reactions using sacrificial agent. Based on experimental results above, we raise an assumption of the photocatalytic mechanism that the photogenerated electrons reacted with nitrogen to form ammonia and the photogenerated holes reacted with water to produce oxygen. To ensure this photocatalytic mechanism, nitrogen fixation reaction was conducted in a closed system with a quartz plate, and the gas in the reaction system after 2 h N_2RR was determined by gas chromatograph (Agilent 7890) with a thermal conductivity detector (TCD). Both CN and CG-CN-3 exhibit oxygen signal at about 1.8 min (Fig. S12), which indicates that the photogenerated holes were involved in water oxidation instead of catalyst self-oxidation. As presented in Fig. 4d, there was no trace of ammonia detected during nitrogen fixation over CG-CN-3 under the Ar saturation, confirming that the nitrogen is vital for the ammonia evolution. Control experiments with using aprotic solvent (CH_3CN) instead of water revealed no ammonia yield, suggesting the indispensability of H_2O as the proton source for the photocatalytic nitrogen fixation (Fig. 4d). To further exclude the influence of ammonia impurity in high-purity N_2 and air atmosphere a series of control experiments without high-purity N_2 , light source and photocatalyst are carried (Fig. S13 and S14). Almost no NH_3 (NH_4^+) generated under the control conditions.

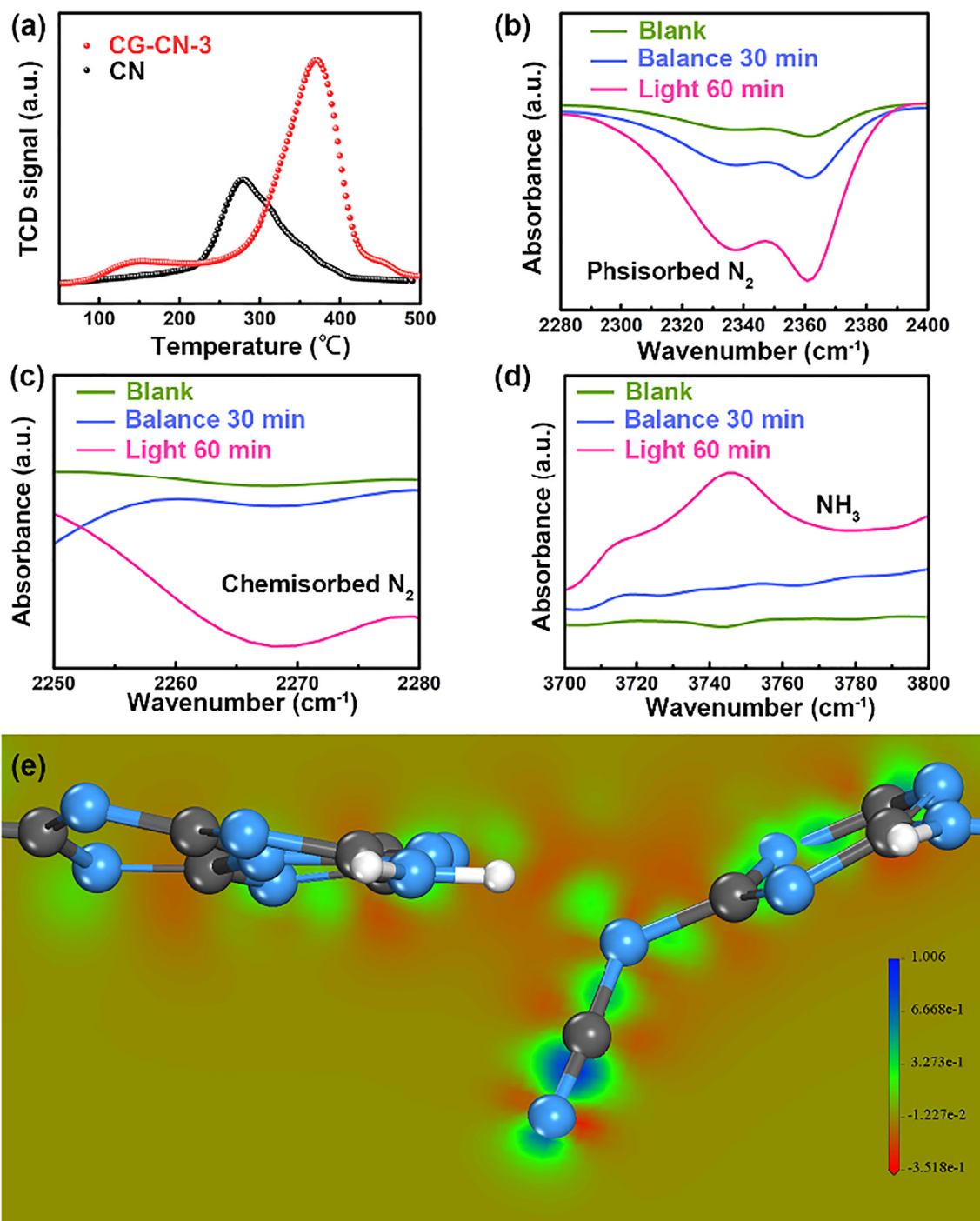


Fig. 5. (a) N₂-TPD of CN and CG-CN-3. (b–d) In situ FT-IR spectra on CG-CN-3. (e) The charge density difference of CG-CN-3.

To eliminate the interference of decomposition of catalyst, the ¹H NMR and ion chromatography measurements are carried (Fig. S15). ¹⁵NH₄⁺ was detected following catalysis in a ¹⁵N₂ saturated solution, which exhibited the same double peak as ¹⁵NH₄⁺ standard solution in ¹H NMR spectra (Fig. S15a). The results verified that the ammonia was generated from the photocatalytic nitrogen fixation rather than from decomposition of catalyst. In addition, ammonia production in solution after 0 min, 30 min, and 60 min N₂RR was determined by ion chromatography (Fig. S15b). The characteristic signal of ammonia accumulated with the reaction time increasing [44], which indicated the ammonia generated from the photocatalytic nitrogen fixation. These results can strongly evidence that the as-detected NH₃ indeed is

obtained from photocatalytic N₂RR process rather than from the environment or decomposition of catalyst.

Nitrogen temperature programmed desorption (N₂-TPD) measurements and in situ Fourier transform infrared were performed to prove the enhancement of N₂ adsorption and activation of the cyano group modified g-C₃N₄. As shown in Fig. 5a, the peaks at 270 °C of CN and 370 °C of CG-CN-3 represent chemical adsorption, and the intensity of the CN peak is much weaker than that of the CG-CN-3. The higher intensity and desorption temperature of N₂-TPD on CG-CN-3 certify the enhanced nitrogen activation by the cyano group. Fig. 5b, c and d displayed changes after injection of N₂ and light 60 min. As shown in Fig. 5b, characteristic absorption bands assigned to the N₂ physical

absorption at 2320–2380 cm^{-1} exhibited an obvious enhancement, which corresponded to the TPD results [45]. Absorption bands of N_2 chemisorption at 2270 cm^{-1} emerged after 60 min radiation under visible light, while no significant change appeared after N_2 saturation in the dark, shown in Fig. 5c [46]. The results support the existence of nitrogen fixation reaction intermediate under visible light. As displayed in Fig. 5d, the specific absorption of NH_3 appeared a synchronous variation with N_2 chemisorption, which further testified the existence of the intermediate during the reaction process [47]. Besides, The feasibility of N_2 activation on g- C_3N_4 with cyano group is verified by theoretical calculation, shown in Fig. 5e. The formation energy of the cyano group bonding to the framework of g- C_3N_4 is -2.86 eV per bond (for organic compounds, the bond energy varies from 1.25 ± 0.11 eV per bond to 5.17 ± 0.13 eV per bond in handbook of bond dissociation energies in organic compounds), suggesting the reasonable bonding between cyano group and g- C_3N_4 . The charge density of the N atom directly connected to cyano group is obviously decreased, while the charge density of $\text{C}\equiv\text{N}$ triple bonds is increased (Fig. 5e). The lower charge density led to an enhanced Lewis acidity, which indicated an improved N_2 activation ability [48].

4. Conclusion

In summary, a series of cyano group modified graphitic carbon nitride is successfully synthesized. The treatment of catalysts by KCl/LiCl molten salt process with melamine introduces cyano group into the framework of g- C_3N_4 . The cyano group can improve the separation of photogenerated electron-hole pairs and simultaneously promote the nitrogen activation due to the strong electron-withdrawing character. CG-CN samples show high visible-light activities in photocatalytic nitrogen fixation with no sacrificial agent. The photocatalytic nitrogen fixation rate of CG-CN-3 ($442.92 \mu\text{g h}^{-1} \text{g}_{\text{cat}}^{-1}$) is 12.8 times as high as that of CN ($34.35 \mu\text{g h}^{-1} \text{g}_{\text{cat}}^{-1}$). This simple molten salt calcination route to prepare cyano defective g- C_3N_4 could be used to introduce cyano group into other nitrogen-based materials for improved solar efficiency.

CRedit authorship contribution statement

Yi Kong: Investigation, Methodology. **Chade Lv:** Investigation, Data curation. **Congmin Zhang:** . **Gang Chen:** Conceptualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsusc.2020.146009>.

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